1.

(1) $C_0(K)$ is free over $\{v_0, v_1, v_2, v_3, v_4\}$ and $C_1(K)$ is free over $e_{ij} = [v_i, v_j]$ for $i = 0, j = 1, 2, 3, 4$ and $(i, j) = (1, 2), (3, 4)$. ∂_1 is given by $e_{ij} \mapsto v_j - v_i$.
(2) $\mathrm{Ker}\partial_1 = \langle e_{01} + e_{12} - e_{02}, e_{03} + e_{34} - e_{04}\rangle$ so $H_1(K) = \mathrm{Ker}\partial_1 \cong \mathbb{Z}^2$.
(3) $H_0(K) = C_0(K)/\mathrm{Im}(K) = C_0(K)/\langle v_i - v_0\rangle \cong \mathbb{Z}$ so K is path-connected.

2.

(1) $C_0(K, K_0) = \langle v_3\rangle$, $C_1(K, K_0) = \langle e_{02}, e_{03}, e_{23}\rangle$ where $e_{ij} = [v_i, v_j]$, and $C_2(K, K_0) = \langle f_1, f_2\rangle$ where f_1, f_2 are the two triangles.
The boundary maps are given by $\partial_1 : e_{02} \mapsto 0, e_{03} \mapsto v_3, e_{23} \mapsto v_3$ and $\partial_2 : f_1 \mapsto -e_{02}, f_2 \mapsto e_{23} - e_{03} + e_{02}$.
(2) $\mathrm{Ker}\partial_1 = \langle e_{02}, e_{03} - e_{23}\rangle$, $\mathrm{Im}\partial_1 = \langle e_{02}, e_{03} - e_{23}\rangle$ and $\mathrm{Ker}\partial_2 = 0$.
(3) By (2), $H_n(K, K_0) = 0 \forall n$.

3.

Trivial since the cylinder deformation contracts to its bottom.
(1) $C_0(K, K_0) = \langle w_0, w_1, w_2\rangle$, $C_1(K, K_0) = \langle [u_i, w_i], [u_i, u_j], [u_i, w_{i+1}]\rangle$, $C_2(K, K_0)$ $= \langle [u_i, u_{i+1}, w_i], [u_i, w_{i-1}, w_i]\rangle$. By definition the edges in K_0 are sent to zero.
(2) $\mathrm{Ker}\partial_1 = \langle [u_i, u_j], [u_i, w_i] - [u_{i-1}, w_i]\rangle$ and likewise $\mathrm{Im}\partial_2 = \langle [u_i, u_j], [u_i, w_i] - [u_{i-1}, w_i]\rangle = \mathrm{Ker}\partial_1$.
(3) Clearly $\mathrm{Im}\partial_1 = C_0$ and $\mathrm{Ker}\partial_2 = 0$ so $H_n(K, K_0) = 0 \forall n$.

4.

Consider the CW structure as one 0-cell v, one 1-cell e and one 2-cell f, with boundary maps $\partial_1 : e \mapsto 0$ and $\partial_2 : f \mapsto 2e$, then $H_0(K) = \mathbb{Z}, H_1(K) = \mathbb{Z}/2\mathbb{Z}, H_2(K) = 0$.

5.

(1) $\partial z = \sum_\sigma \partial\sigma$, so the newly added edges cancel out, leaving the eight outer edges, which also cancel each other out. $\partial \sum n_\sigma \sigma = 0$ forces the coefficients of neighboring faces to agree, so it must be in $\langle z\rangle$, and hence $H_2(K) = \langle z\rangle = \mathbb{Z}$.
(2) Lift the 1-cycle c to the octagon as $\tilde{c}$, then its boundary consists of vertices of the octagon, so we can find an edge chain b homologous to $\tilde{c}$. Glueing back to K we infer c is homologous to the sum of a_i, b_i.
(3) Suppose any composition of a_i, b_i bounds a 2-cycle, cut open K and apply (1) to infer every triangle in the octagon to have the same coefficient. Hence when glueing back, the boundary of the triangles cancel out, therefore the coefficients of a_i, b_i must be zero.
Therefore. $H_0(K) = \mathbb{Z}, H_1(K) = \mathbb{Z}^4$.

6.

(1) The 3x3 square, which form a torus, has homology groups $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^2$ and $H_2 = \mathbb{Z}$.
(2) For the group $G = \langle x_1, \cdots, x_n \mid r_1, \cdots, r_m \rangle$, $C_0 = \langle v \rangle$, $C_1 = \langle x_1, \cdots, x_n \rangle$ and $C_2 = \langle r_1, \cdots, r_m \rangle$, with boundary maps $\partial_1 = 0$ and $\partial_2 : r_i = \prod x_{i,j}^{\varepsilon_{i,j}} \mapsto \sum \varepsilon_{i,j} x_{i,j}$.
Hence $H_0 = \mathbb{Z}$, $\mathrm{Ker}\partial_1 = \mathbb{Z}^n$, $\mathrm{Im}\partial_2 = \langle \sum \varepsilon_{i,j} x_{i,j} \rangle$, and $\mathrm{Ker}\partial_2 = \{(t_1, \cdots, t_m : \sum t_i \varepsilon_{i,j} x_{i,j} = 0)\}$.
Therefore $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^n / \langle \sum \varepsilon_{i,j} x_{i,j} \rangle = G^{\mathrm{ab}}$ and $H_2 = \{(t_1, \cdots, t_m) \in \mathbb{Z}^m : \sum t_i \varepsilon_{i,j} x_{i,j} = 0\}$.
(3) For $G = \langle a, b, c \mid ab^{-1}c, a^2bc^{-1}, b^3c^{-2} \rangle$, we have $H_1 = G^{\mathrm{ab}} = \mathbb{Z}/3\mathbb{Z}$, and $H_2 = 0$.